ADITYA UNIVERSITY

TIME, SPEED & DISTANCE

P. Vijay Kumar

Aptitude Trainer

Department of Training & Placement

Aditya University

ADITYA UNIVERSITY

Introduction

Time, speed, and distance form the basis of motion, describing how fast an object covers space. The fundamental relationship is Distance = Speed x Time. Speed is the rate of motion (Distance/Time), distance is space covered, and time is the duration, often featuring inverse proportionality where higher speed equals lower time.

ADITYA UNIVERSITY

Concept and Definitions

Speed (S): Distance / Time.

Average Speed : Total Distance / Total Time.

Relative Speed : Opposite Directions: Speeds are added (S1 + S2).Same Direction: Speeds are subtracted (S1 – S2)

Unit Conversions :kmph → mps : Multiply with 5/18 and Vice -Versa.

ADITYA UNIVERSITY

Common Problem Types

Basic Time, speed and distance questions based on Average speed and Relative Speed.

Trains: Calculating length, speed, or time when crossing poles, platforms, or other trains.

Boats and Streams: Calculating upstream (against flow) and downstream (with flow) speeds.

Races: Problems related to linear or circular motion.

Questions

Ques.1 A person crosses a 600 m long street in 5 minutes. What is his speed in km per hour?

a) 11

b) 7.2

c) 8.4

d) 10

ADITYA UNIVERSITY

Questions

Ques.2 An airplane covers a certain distance at a speed of 240 kmph in 5 hours. To cover the same distance in 1(2/3)hours, it must travel at a speed of

a) 300 kmph

b) 360 kmph

c) 400 kmph

d) None of these

ADITYA UNIVERSITY

Questions

Ques.3 If a person walks at 14 km/hr instead of 10 km/hr, he would have walked 20 km more. The actual distance travelled by him is

a) 50 km

b) 37 km

c) 40 km

d) None of these

ADITYA UNIVERSITY

Questions

Ques.4 Excluding stoppages, the speed of a bus is 54 kmph and including stoppages, it is 45 kmph. For how many minutes does the bus stop per hour?

a) 4

b) 9

c) 10

d) None of these

ADITYA UNIVERSITY

Questions

Ques.5 A man complete a journey in 10 hours. He travels first half of the journey at the rate of 21 km/hr and second half at the rate of 24 km/hr. Find the total journey in km.

a) 220 km

b) 224 km

c) 230 km

d) 234 km

ADITYA UNIVERSITY

Questions

Ques.6 A man on tour travels first 160 km at 64 km/hr and the next 160 km at 80 km/hr. The average speed for the first 320 km of the tour is:

a) 35.55 kmph

b) 36 kmph

c) 71.11 kmph

d) 71 kmph

ADITYA UNIVERSITY

Questions

Ques.7 Robert is travelling on his cycle and has calculated to reach point A at 2 P.M. If he travels at 10 kmph, he will reach there at 12 noon if he travels at 15 kmph. At what speed(in kmph) must he travel to reach A at 1 P.M.?

a) 14

b) 8

c) 12

d) 10

ADITYA UNIVERSITY

Questions

Ques.8 **Two boys starts from the same place walking at the rate of 5 kmph and 5.5 kmph respectively in the same direction. What time will they take to be 8.5 km apart?**

a) 14 hr

b) 18 hr

c) 20 hr

d) 17 hr

ADITYA UNIVERSITY

Questions

Ques.9 A Person is moving with speed of 16kmph. A truck coming behind crosses him. Person can see the truck upto 3 min. and now the distance between the truck and person is 1400m. Find the speed of the truck.

a) 44 kmph

b) 40 kmph

c) 8 kmph

d) None of these

ADITYA UNIVERSITY

Questions

Ques.10 : A bus travels first 240 km @80kmph and next 360km @90kmph. What is the average speed ?

a) 85.71

b) 88

c) 90.99

d) None of these

Questions

Ques11.A delivery driver travels from Town A to Town B at an average speed of 40 km/h. On the return trip from Town B to Town A, taking the exact same route, the driver increases their speed to 60 km/h. What is the average speed for the entire round trip?

a) 48 kmph

b) 50 kmph

c) 52 kmph

d) 45 kmph

Questions

Ques12. An athlete runs the first 10 km of a race in 40 minutes. For the next 20 km, they maintain a constant speed of 12 km/h. Finally, they sprint the remaining 5 km in 15 minutes. What is the athlete's average speed for the entire 35 km race?

a) 10.5 kmph

b) 11.2 kmph

c) 12.0 kmph

d) 13.5 kmph

Questions

ADITYA UNIVERSITY

Ques13. The distance between City A and City B is 440 km. At 4:00 PM, a train leaves City A heading toward City B at an average speed of 40 km/h. One hour later, at 5:00 PM, another train leaves City B heading toward City A at an average speed of 60 km/h. At what time will the two trains meet?

a) 8pm

b) 8:30pm

c) 9pm

d) 0:30pm

ADITYA UNIVERSITY

Thank You